Lesson 8

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OPERATING SYSTEM

MANAGING, COORDINATING & MONITORING RESOURCES



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WHAT IS AN OPERATING SYSTEM?

An **Operating System (OS)** is a set of programs that coordinate all the activities among computer or mobile device hardware.

Lesson 8 HOW AN OPERATING SYSTEM WORKS

- Start and shut down a computer or mobile device.
- Provide a user interface.
- Manage programs.
- Manage Memory
- Coordinate tasks
- Configure devices
- Monitor Performance
- Establish Internet Connection

Lesson 8 HOW AN OPERATING SYSTEM WORKS

- Provide file management and other device or media-related tasks
- Updating operating system software
- Control a network
- Administer security


administer security
start and shut down the computer
provide a user interface
control a network
manage programs
update automatically
operating system functions
manage memory
provide file management and other tools
coordinate tasks and configure devices
establish an internet connection
monitor performance

- **Starting Computers and Mobile Devices**
  - If a computer or mobile device is off, you press apower button to turn it on.



- An operating system includes various power options

Sleep mode saves any open documents running programs or apps to RAM, turns off all unneeded functions, and then places the computer in alow-power state

Hibernate mode saves any open documents and running programs or apps to an internal hard drive before removing power from the computer or device.

- A **user interface (UI)** controls how you enter data and instructions and how information is displayed on the screen.
- With a **graphical user interface (GUI)**, you interact with menus and visual images.



- In a **command-line interface (CLI)**, a user types commands represented by short keywords or abbreviations or presses special keys on the keyboard to enter data and instructions.

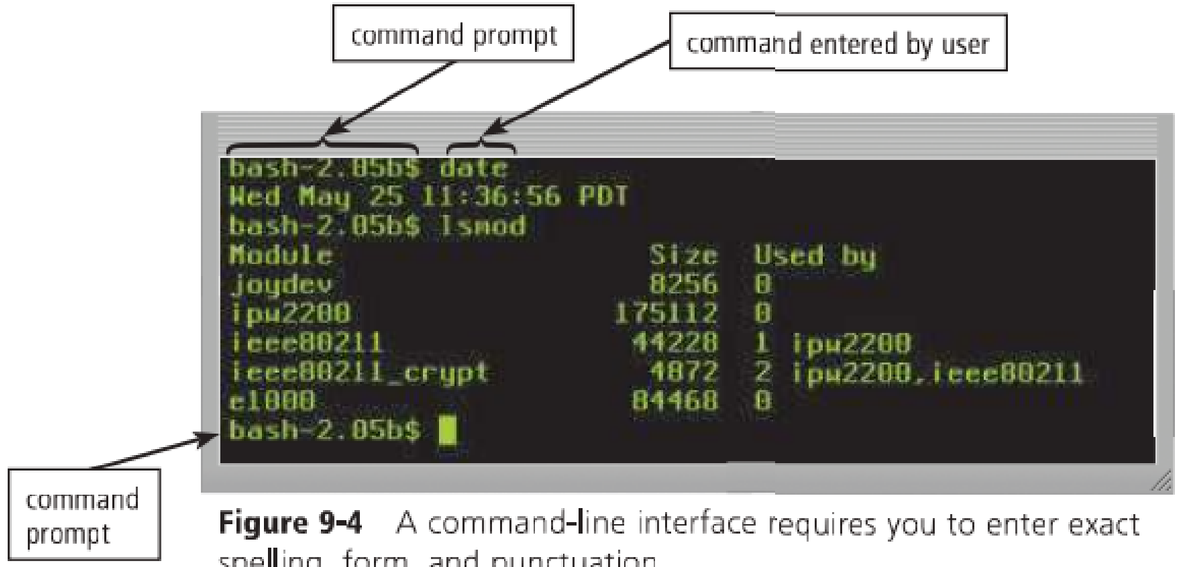


Figure 9-4 A command-line interface requires you to enter exact spelling, form, and punctuation.

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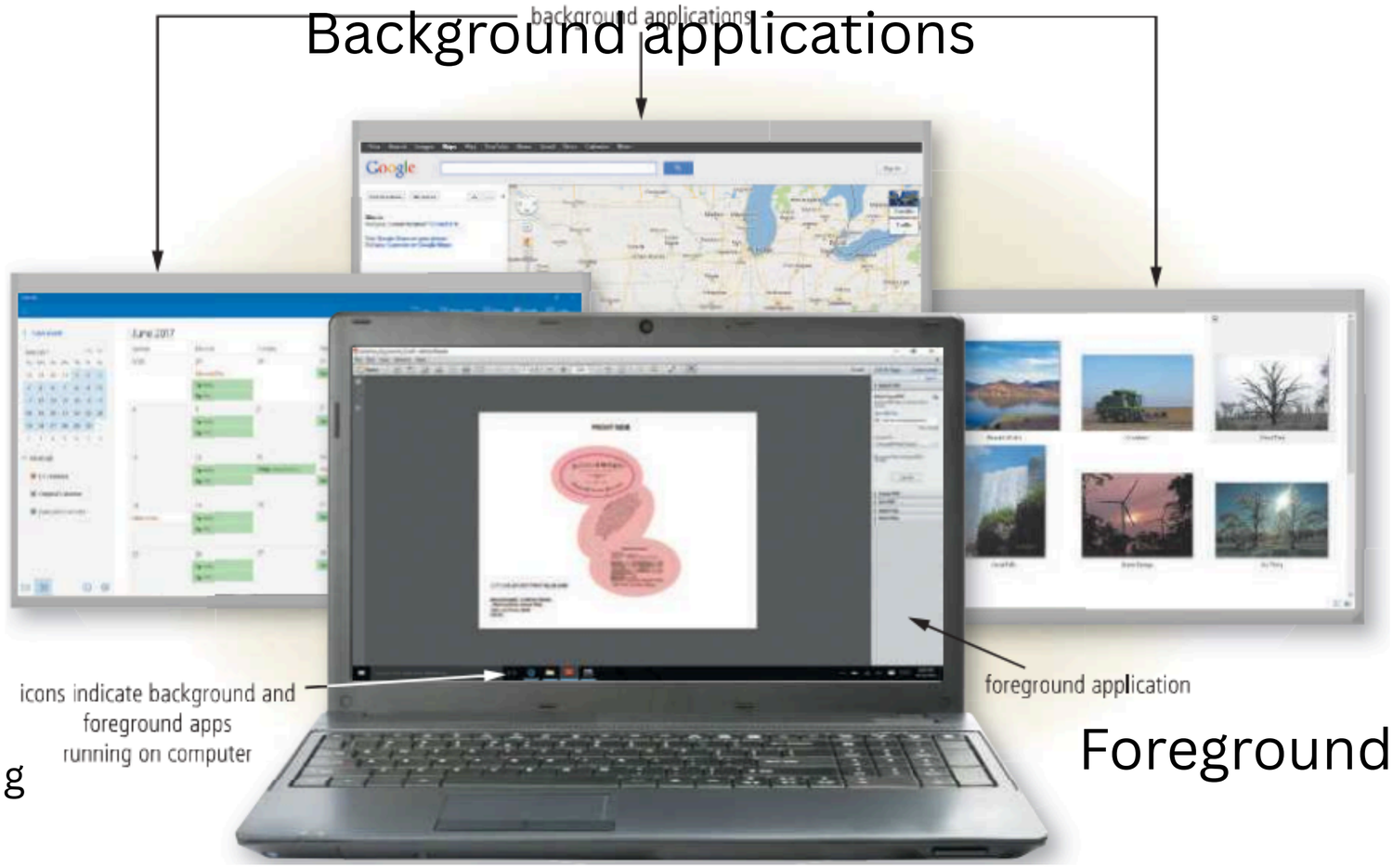
- How an operating system handles programs directly affects your productivity

Single tasking and multitasking

Foreground and background

Single user and multiuser

Background applications

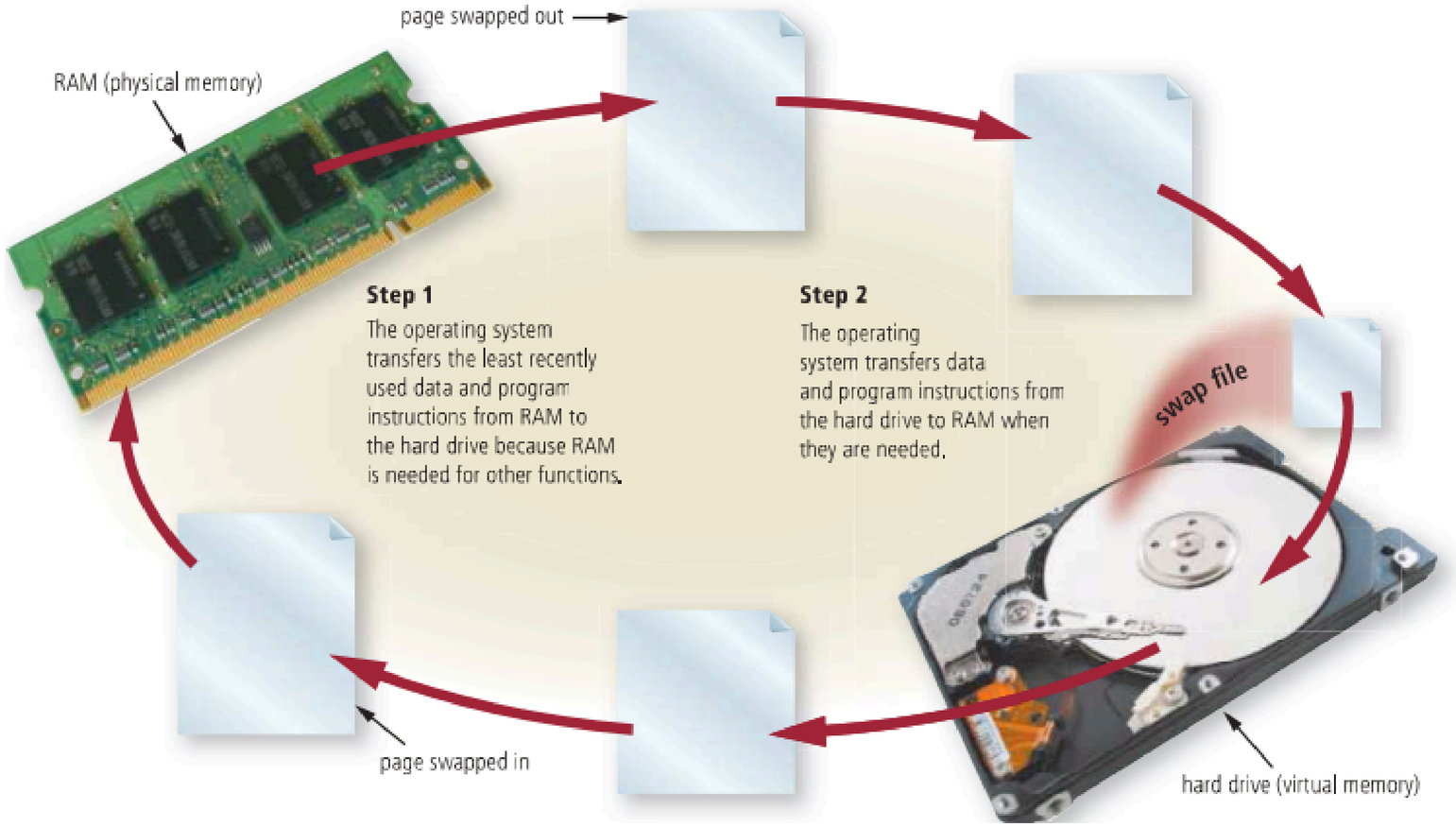


icons indicate background and foreground apps running on computer

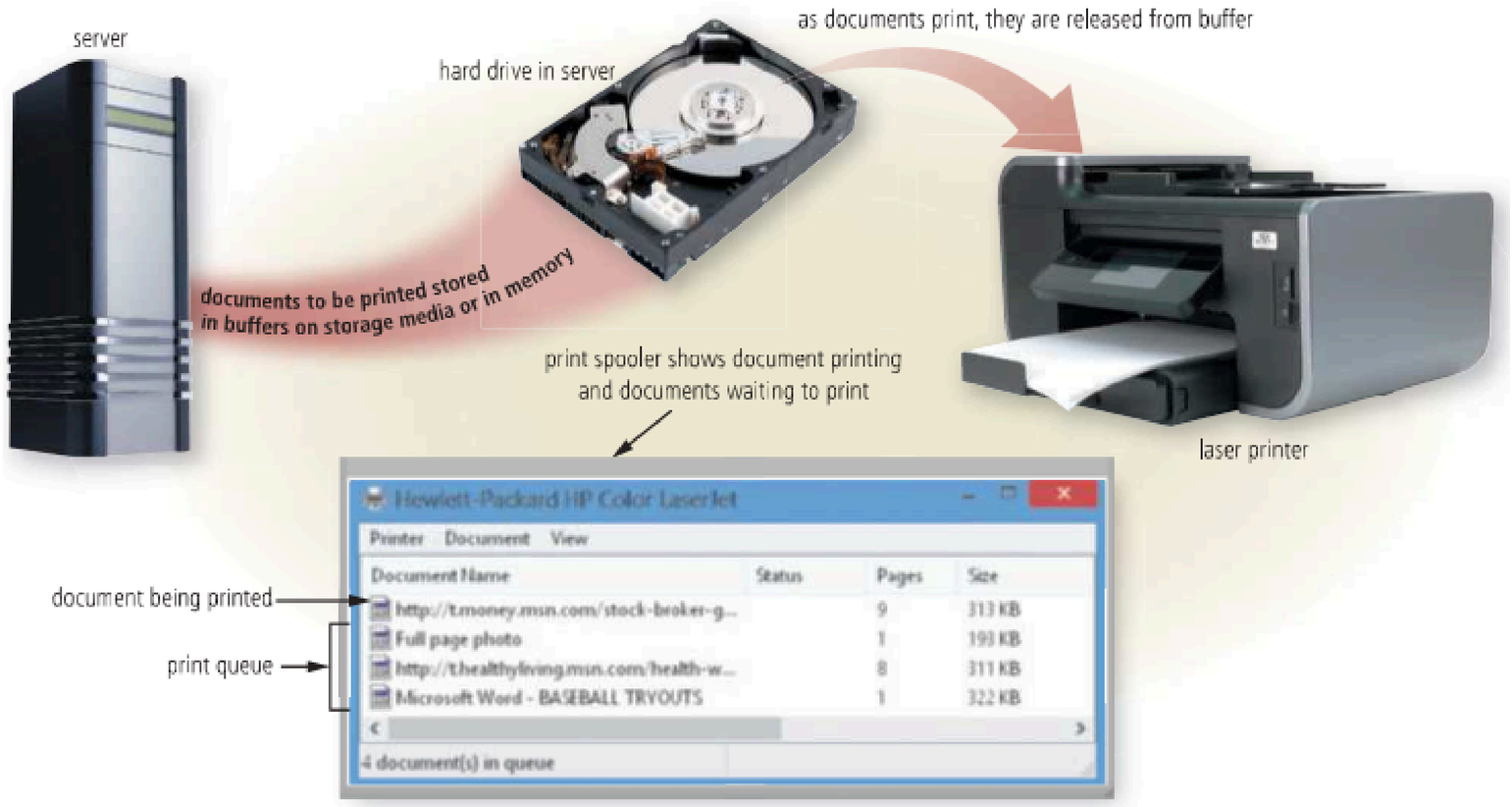
Foreground applications

- **Memory management** optimizes the use of the computer or device’s internal memory.
- Virtual memory is a portion of a storage medium functioning as additional RAM.

How a Computer Might Use Virtual Memory

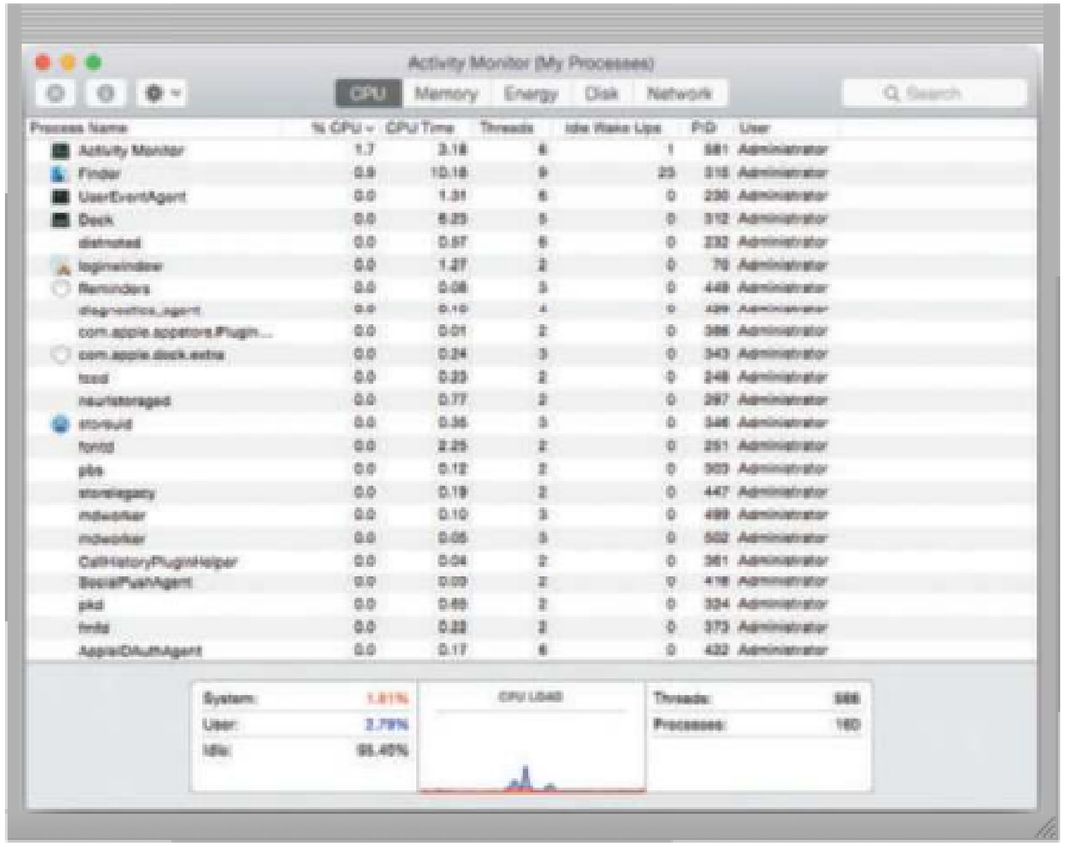


- The operating system determines the order in which tasks are processed.

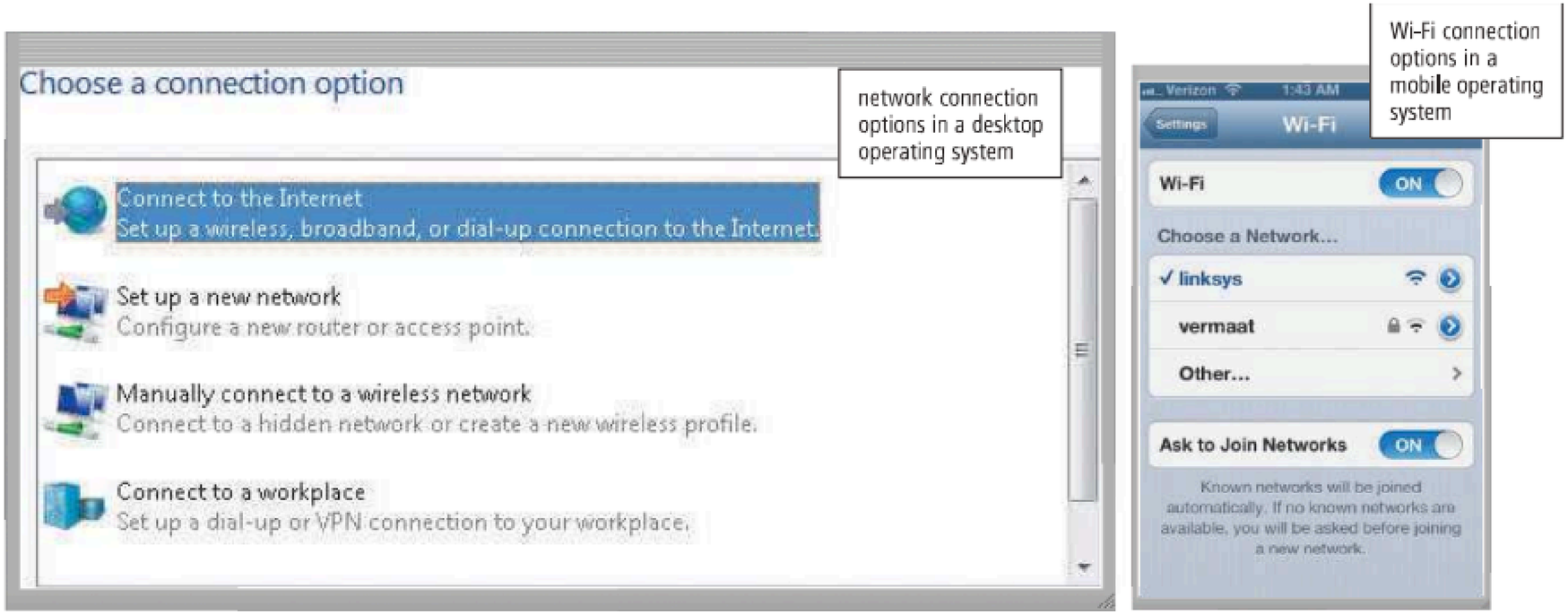


- **Configure Devices**
  - A driver is a small program that tells the operating system how to communicate with a specific device
  - Plug and Play automatically configures new devices as you install or connect them.

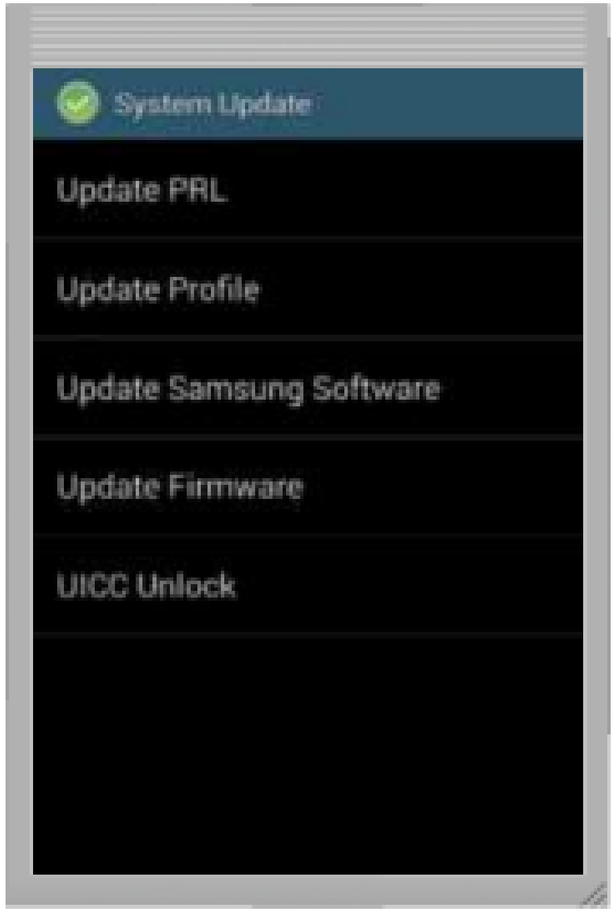
- **Performance Monitor**
  - is a program that assesses and reports information about various computer resources and devices.



- Operating systems typically provide a means to establish internet connections



- Many programs, including operating systems, include an **automatic update** feature that regularly provides new features or corrections to the program.



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- Operating systems often provide users with a variety of tools related to maging a computer, its devices, or its programs.

File Manager

Search

Image Viewer

Uninstaller

Disk Cleanup

Disk Defragmenter

Screen Saver

File Compression

PC Maintenance

Backup and Restore

Table 9-1 File, Disk, and System Management Tools

Tool	Function
File Manager	Performs functions related to displaying files; organizing files in folders; and copying, renaming, deleting, moving, and sorting files
Search	Attempts to locate files on your computer or mobile device based on specified criteria
Image Viewer	Displays, copies, and prints the contents of graphics files
Uninstaller	Removes a program or app, as well as any associated entries in the system files
Disk Cleanup	Searches for and removes unnecessary files
Disk Defragmenter	Reorganizes the files and unused space on a computer's hard disk so that the operating system accesses data more quickly and programs and apps run faster
Screen Saver	Causes a display's screen to show a moving image or blank screen if no keyboard or mouse activity occurs for a specified time
File Compression	Shrinks the size of a file(s)
PC Maintenance	Identifies and fixes operating system problems, detects and repairs drive problems, and includes the capability of improving a computer's performance
Backup and Restore	Copies selected files or the contents of an entire storage medium to another storage location

- Some operating systems are designed to work with a server on a network.
- These multiuser operating system allow multiple users to share a printer

- A network administrator uses the server operating system to:
  - add and remove users, computers and other devices
  - configure the network, install software and administer network security



- A user account enables a user to **sign in** to, or access resources on, a network or computer.
  - A **user name**, or a user ID, identifies a specific user
  - A **password** is a private combination of characters associated with the user name.

Types of Operating Systems:

- Desktop Operating System
- Server Operating System
- Mobile Operating System

DESKTOP OPERATING SYSTEM

- A desktop operating system is a complete operating system that works on desktops, laptops, and some tablets.

Windows

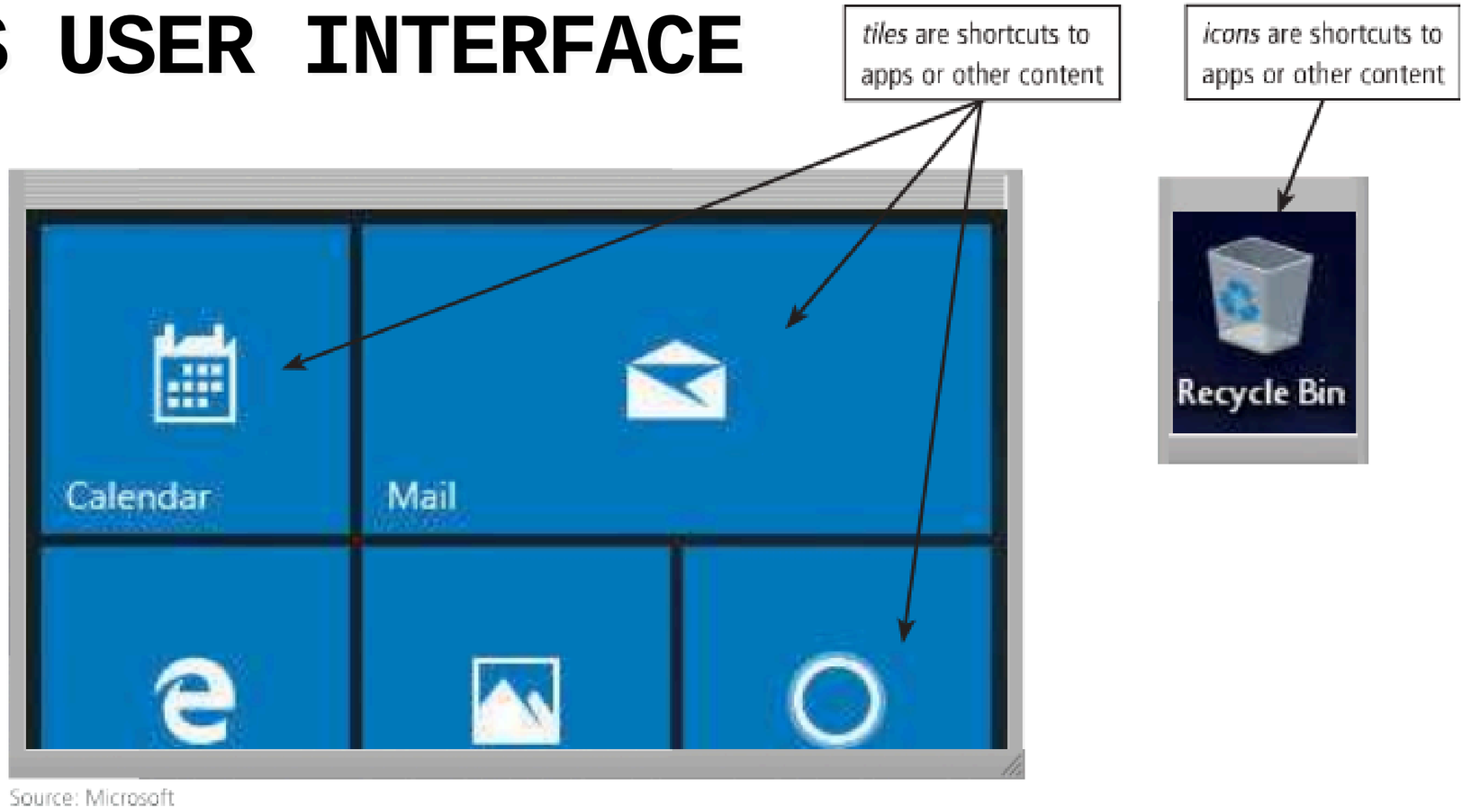
Mac OS

UNIX

Linux

Chrome OS

WINDOWS OS USER INTERFACE



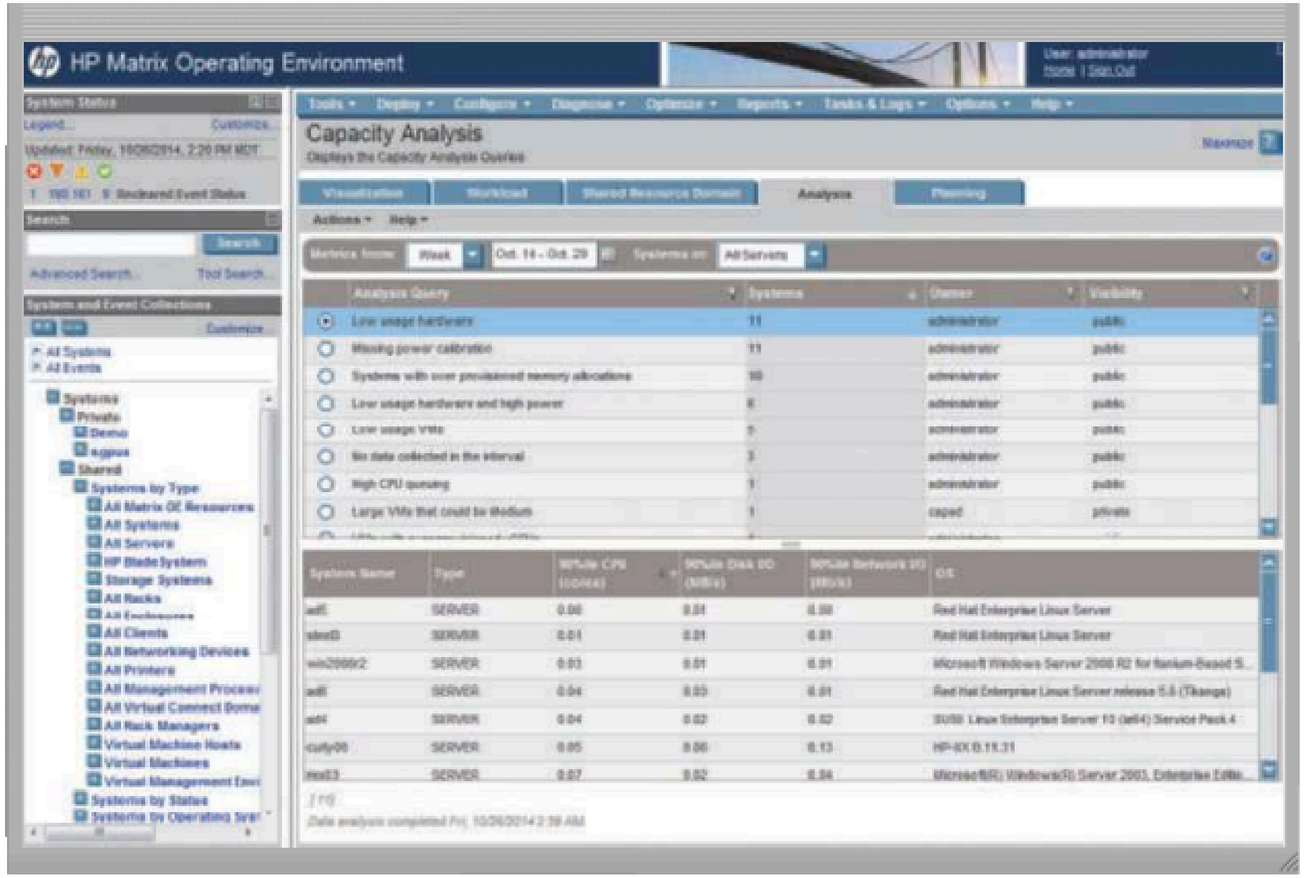
Source: Microsoft

MAC OS USER INTERFACE



UNIX

- is a multitasking operating system developed in the early 1970s.



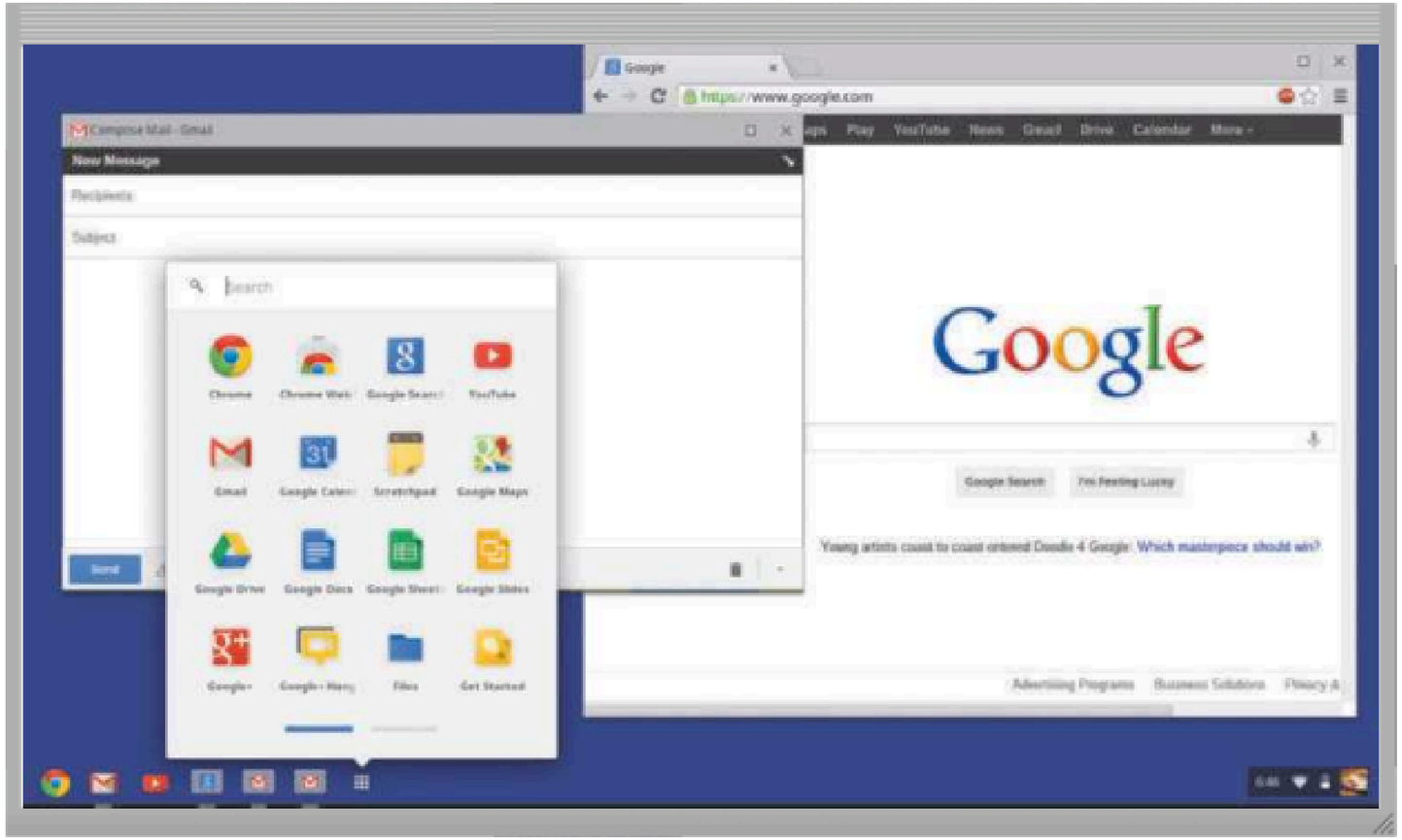
LINUX

- is a popular, multitasking UNIX-based operating system



CHROME OS

- is a linux-based operating system designed to work primarily with web apps



SERVER OPERATING SYSTEM

- A server system is a multiuser operating system that organizes and coordinates how multiple users access and share resources on a network.

Windows Server

UNIX

Linux

OS X Server

Windows Server

- Windows server enables organizations to manage applications and websites on-site and/or on the cloud.

OS X Server

- OS X Servers enables organizations to collaborate, share files, host websites and mail servers, and more on Mac computers and iOS devices.

UNIX

- Capable of handling a high volume of transactions in a multiuser environment and working with multiple processors.

Linux

- Because it provides a secure, stable multiuser environment, Linux often is used on web servers and on supercomputers

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MOBILE OPERATING SYSTEMS

- The operating system on mobile devices and many consumer electronics is called amobile operating system and resides on firmware.

Android

iOS

Windows Phone

ANDROID

- Android is an open source, Linux-based mobile operating system designed by Google for smartphones and tablets



iOS

- developed by Apple, is a propriety mobile operating system, specifically made for Apple’s mobile devices.



Windows Phone

- developed by Microsoft, is a propriety mobile operating system that runs on some smartphones.



MANAGEMENT OF FILES AND FOLDERS

- The File Manager (or File Management System) is the manager in the Operating System that creates the illusion that there are files and folders being stored in computer memory.

VIRTUAL MACHINES

- A virtual machine is a software-based computer that runs within a physical computer (known as the host). It emulates the behavior of a physical machine and can run its own operating system (guest OS) and applications independently just like a real computer.
- Example: you can run a Linux VM inside a Windows PC using tools like VMware

LICENSING AN OPERATING SYSTEM

- Licensing an operating system involves understanding the legal framework governing its use, distribution, and modification.
- Operating systems manage software licensing through various mechanisms, ensuring compliance with legal requirements and preventing unauthorized use.
- Proper licensing is crucial for legal compliance and access to updates and support from the software vendor.

SECURING THE OPERATING SYSTEM

- Installing only necessary software
- configuring users and permissions properly
- Applying patches and updats
- Performing regular security monitoring
- Backups and testing

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THANK YOU